

Tutorial-6 (Discrete Random Variable)

Problem 1

Let X be a discrete random variable with the following PMF:

$$P_X(x) = \begin{cases} \frac{1}{2} & \text{for } x = 0 \\ \frac{1}{3} & \text{for } x = 1 \\ \frac{1}{6} & \text{for } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find R_X , the range of the random variable X .
- (b) Find $P(X \leq 1.5)$.
- (c) Find $P(0 < X < 2)$.
- (d) Find $P(X = 0 \mid X < 2)$.

Problem 2

I roll two dice and observe two numbers X and Y . If $Z = X - Y$, find the range and PMF of Z .

Problem 3

50 students live in a dormitory. The parking lot has the capacity for 30 cars. If each student has a car with probability $\frac{1}{2}$ (independently from other students), what is the probability that there won't be enough parking spaces for all the cars?

Problem 4

For each of the following random variables, find $P(X > 5)$, $P(2 < X \leq 6)$, and $P(X > 5 \mid X < 8)$. You do not need to provide the numerical values for your answers. In other words, you can leave your answers in the form of sums.

- (a) $X \sim \text{Geometric}(\frac{1}{5})$
- (b) $X \sim \text{Binomial}(10, \frac{1}{3})$
- (c) $X \sim \text{Pascal}(3, \frac{1}{2})$
- (d) $X \sim \text{Hypergeometric}(10, 10, 12)$
- (e) $X \sim \text{Poisson}(5)$

Problem 5

In this problem, we would like to show that the geometric random variable is memoryless. Let $X \sim \text{Geometric}(p)$. Show that

$$P(X > m + l \mid X > m) = P(X > l) \quad \text{for } m, l \in \{1, 2, 3, \dots\}$$

We can interpret this in the following way: remember that a geometric random variable can be obtained by tossing a coin repeatedly until observing the first heads. If we toss the coin several times and do not observe a heads, from now on it is as if we start all over again. In other words, the failed coin tosses do not impact the distribution of waiting time from now on. The reason for this is that the coin tosses are independent.

Problem 6

The number of emails that I get on a weekday (Monday through Friday) can be modeled by a Poisson distribution with an average of $\frac{1}{6}$ emails per minute. The number of emails that I receive on weekends (Saturday and Sunday) can be modeled by a Poisson distribution with an average of $\frac{1}{30}$ emails per minute.

- (1) What is the probability that I get no emails in an interval of length 4 hours on a Sunday?
- (2) A random day is chosen (all days of the week are equally likely to be selected), and a random interval of length one hour is selected on the chosen day. It is observed that I did not receive any emails in that interval. What is the probability that the chosen day is a weekday?

Problem 7

Let X be a discrete random variable with the following CDF:

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{6} & \text{for } 0 \leq x < 1 \\ \frac{1}{2} & \text{for } 1 \leq x < 2 \\ \frac{3}{4} & \text{for } 2 \leq x < 3 \\ 1 & \text{for } x \geq 3 \end{cases}$$

Find the range and PMF of X .

Problem 8

Let $X \sim \text{Geometric}\left(\frac{1}{3}\right)$ and let $Y = |X - 5|$. Find the range and PMF of Y .

Problem 9

Let $X \sim \text{Geometric}(p)$. Find $\text{Var}(X)$.

Problem 10

Suppose that $Y = -2X + 3$. If we know $E[Y] = 1$ and $E[Y^2] = 9$, find $E[X]$ and $\text{Var}(X)$.

Problem 11

Coupon collector's problem: Suppose that there are N different types of coupons. Each time you get a coupon, it is equally likely to be any of the N possible types. Let X be the number of coupons you will need to get before having observed each coupon at least once.

- (a) Show that you can write $X = X_0 + X_1 + \dots + X_{N-1}$, where $X_i \sim \text{Geometric}\left(\frac{N-i}{N}\right)$.
- (b) Find $E[X]$.

Problem 12

Let X be a random variable with mean $E[X] = \mu$. Define the function $f(\beta)$ as

$$f(\beta) = E[(X - \beta)^2].$$

Find the value of β that minimizes $f(\beta)$.